

JavaScript Basics — Assignment 04

Question 1 : JavaScript Operators

// Two number variables

```
let num1 = 20;
```

```
let num2 = 5;
```

// Arithmetic Operators

```
console.log("Arithmetic Operators:");
```

```
console.log("Addition:", num1 + num2);
```

```
console.log("Subtraction:", num1 - num2);
```

```
console.log("Multiplication:", num1 * num2);
```

```
console.log("Division:", num1 / num2);
```

```
console.log("Modulus:", num1 % num2);
```

// Increment

```
let incrementNum = num1;
```

```
incrementNum++;
```

```
console.log("Increment:", incrementNum);
```

// Decrement

```
let decrementNum = num1;
```

```
decrementNum--;
```

```
console.log("Decrement:", decrementNum);
```

// Assignment Operators

```
console.log("Assignment Operators:");
```

```
let value = 10;
```

```
console.log("Original value:", value);
```

```
value += 5;
```

```
console.log("After += 5:", value);
```

```
value -= 3;
```

```

console.log("After -= 3:", value);

value *= 2;

console.log("After *= 2:", value);

value /= 4;

console.log("After /= 4:", value);

// Comparison Operators

console.log("Comparison Operators:");

console.log("num1 == num2:", num1 == num2);

console.log("num1 === num2:", num1 === num2);

console.log("num1 != num2:", num1 != num2);

console.log("num1 !== num2:", num1 !== num2);

console.log("num1 > num2:", num1 > num2);

console.log("num1 < num2:", num1 < num2);

console.log("num1 >= num2:", num1 >= num2);

console.log("num1 <= num2:", num1 <= num2);

```

Output :

Arithmetic Operators:	question1.js:8
Addition: 25	question1.js:10
Subtraction: 15	question1.js:11
Multiplication: 100	question1.js:12
Division: 4	question1.js:13
Modulus: 0	question1.js:14
Increment: 21	question1.js:19
Decrement: 19	question1.js:24
Assignment Operators:	question1.js:28
Original value: 10	question1.js:32
After += 5: 15	question1.js:35
After -= 3: 12	question1.js:38
After *= 2: 24	question1.js:41
After /= 4: 6	question1.js:44
Comparison Operators:	question1.js:48
num1 == num2: false	question1.js:50
num1 === num2: false	question1.js:51
num1 != num2: true	question1.js:52
num1 !== num2: true	question1.js:53
num1 > num2: true	question1.js:54
num1 < num2: false	question1.js:55
num1 >= num2: true	question1.js:56
num1 <= num2: false	question1.js:57

Question 2 : Student Grade Calculator Using if / else if / else

```
let marks = 85;

let grade;

// Check the marks and determine the grade

if (marks >= 90 && marks <= 100) {

    grade = "A";

}

else if (marks >= 80 && marks <= 89) {

    grade = "B";

}

else if (marks >= 70 && marks <= 79) {

    grade = "C";

}

else if (marks >= 60 && marks <= 69) {

    grade = "D";

}

else {

    grade = "F";

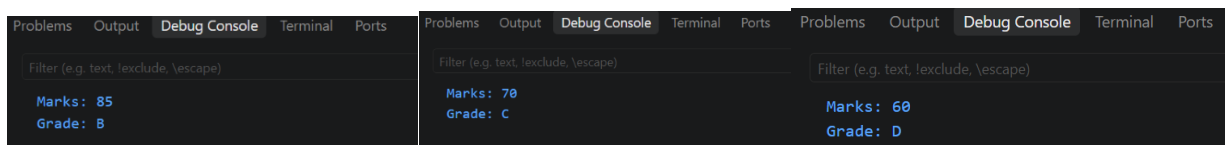
}

// Display marks and grade

console.log("Marks:", marks);

console.log("Grade:", grade);
```

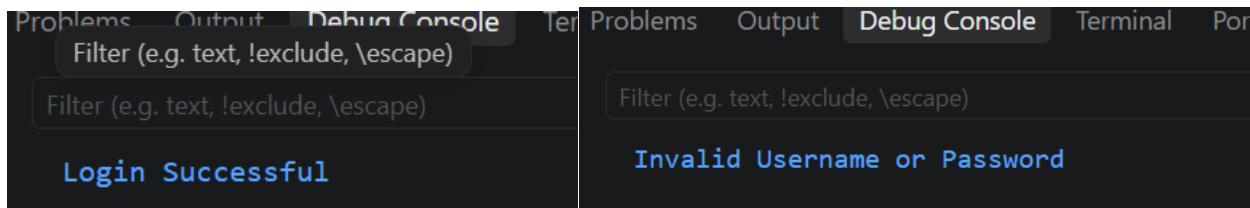
Output :



Question 3: Login Validation Using if / else

```
// Entered username and password
let username = "Fatima";
let password = "12345";
// Correct username and password
let correctUsername = "Fatima";
let correctPassword = "12345";
// Check login details
if (username === correctUsername && password === correctPassword) {
    console.log("Login Successful");
}
else {
    console.log("Invalid Username or Password");
}
```

Output:



Question 4: Ternary Operator:

```
let age = 18;
let result = age >= 18 ? "Eligible to Vote" : "Not Eligible to Vote";
console.log("Age:", age);
console.log(result);
```

Output:



Question5 : Menu Selection Using switch

```
let choice = 1;

switch (choice) {

  case 1:

    console.log("You selected Home");

    break;

  case 2:

    console.log("You selected About");

    break;

  case 3:

    console.log("You selected Services");

    break;

  case 4:

    console.log("You selected Contact");

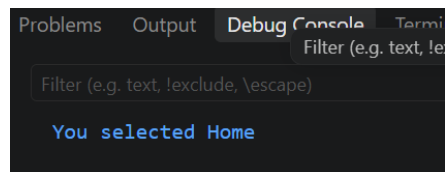
    break;

  default:

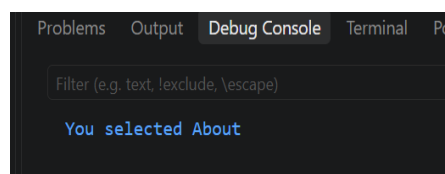
    console.log("Invalid Choice");}
```

Output:

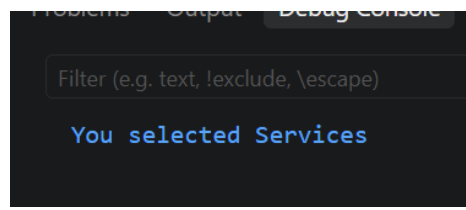
Case1:



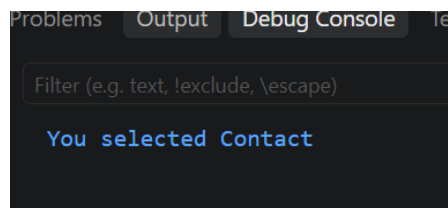
Case2:



Case3:



Case4:



Casedefault:

